

Linear Algebra for AI

Course Metadata

Credit Structure	L-T-P: 3-1-0 Total: 4
Offering Semester	3 rd Sem, Autumn
Level of Subject	UG Level
Compulsory or elective?	Core
Pre-requisite	NIL
Faculty Names	Jiaul H Paik, Adway Mitra, Prabhat Mishra
Overlapping Subjects	MA20105, MA20107, MA30003 (20%), AI61003 (55%)
Justification of overlap	It will be a core subject for BTech in AI program

Course Description

Linear algebra, optimization techniques and statistical methods together form essential tools for most of the algorithms in artificial intelligence and machine learning. In this course, we propose to build some background in these mathematical foundations and prepare students to connect Linear Algebra principles to various AI and ML concepts. This provides an impetus to bridge Linear Algebra to AI/ML thereby preparing students to make this connection when they study advanced AI/ML subjects. The tutorial sessions will focus on computational techniques and applications of linear algebra concepts on machine learning tasks.

Course Content

- Introduction, Norms, distance and linear functions
- Special matrices for ML applications
- Matrix factorization, Low rank matrix approximation and structured low rank approximation
- Convex optimization with gradient descent and accelerated gradient descent
- Missing value and matrix completion methods
- Randomized linear algebra for large matrices
- Introduction to tensors

Lecture-wise Breakup

Sl No.	Module	Unit	Module Hr (48)
1	Introduction	Motivation, Connection to Machine Learning and Deep Learning, Case studies: PageRank algorithm, Image deblurring, Clustering	4
2	Linear Algebra Basics	Norm and distance, Linear and Affine functions and its connection to neural nets, Taylor's Approximation, System of linear equations,	6

		Matrix-matrix multiplication and composition of linear functions, Eigenvalue and Eigenvectors of matrices, Applications in ML such as dimensionality reduction and feature extraction	
3	Special Matrices	Geometric Transformations, Selectors, Incidence Matrix, Convolution. Applications: Image Cropping, Image Blurring, Time series smoothing	3
4	Matrix Factorization	Unconstrained Matrix Factorization, nonnegative Matrix Factorization, Weighted Matrix Factorization, Nonlinear Matrix Factorizations. Applications: Recommender systems, link predictions on graph and word embedding	6
5	Least square	Least squares solution, Constrained least squares. Applications in ML such as optimizing linear regression, and Ridge Regression.	5
6	Low Rank Matrix Approximation	Low rank approximation and structured low rank approximation, SVD and Truncated SVD, Hankel structured low rank approximation. Applications: Dimensionality Reduction, noise removal, outlier detection, feature whitening	6
7	Convex Optimization	Convex optimality conditions: differentiable and non-differentiable cases, Gradient descent, Projected gradient methods, Accelerated gradient methods, alternating minimization. Applications in ML such as optimizing Linear and Logistic Regression.	6
8	Matrix Completion	Missing data and matrix completion problem, Matrix completion problem using nuclear norm. Applications in ML such as blind source separation, data compression, and knowledge discovery.	6
9	Randomized Linear Algebra	Random projections, Norm-squared sampling, Randomized matrix factorization, Randomized matrix approximation	3
10	Introduction to Tensors	Norm and Rank of a Tensor, Matricized form of Tensor, CP and Tucker decomposition	3

Books

1. **Introduction to Applied Linear Algebra: Vectors, Matrices and Least Squares**, Stephan Boyd, Lieven Vandenberghe, Cambridge University Press, 2018.
2. **Linear Algebra and Learning from Data**, Gilbert Strang, Wellesley Cambridge Press, 2019
3. **Statistical Learning with Sparsity: the LASSO and generalizations**, Trevor Hastie, Robert Tibshirani, Martin Wainwright, CRC Press 2017